PVsyst V8.1.8

PVsyst - Simulation report

Grid-Connected System

Project: soalr_trial

Variant: New simulation variant

Sheds, single array

System power: 78.3 kWp

SOLAR_TRIAL - India

Author

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06/11/26 01:46
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Project summary

Geographical Site
SOLAR_TRIAL
India

Situation
Latitude [redacted]
Longitude
Altitude 20 m
Time zone UTC+5.5

Project settings
Albedo 0.20

Weather data
SOLAR_TRIAL
Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

Sheds, single array

Orientation #1
Fixed plane
Tilt/Azimuth 21 / -4 °

Near Shadings
According to strings : Fast (table)
Electrical effect 100 %

User's needs
Unlimited load (grid)

System information

PV Array
Nb. of modules 135 units
Pnom total 78.3 kWp

Inverters
Nb. of units 0.9 unit
Total power 72 kWac
Pnom ratio 1.09

Results summary

Simulation step Hourly

Produced Energy 104858 kWh/year | Specific production 1339 kWh/kWp/year | Perf. Ratio PR 74.89 %

Table of contents

Project and results summary 2
General parameters, PV Array Characteristics, System losses 3
Near shading definition - Iso-shadings diagram 4
Main results 5
Loss diagram 6
Predef. graphs 7

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General parameters

Grid-Connected System — **Sheds, single array**

Simulation step
Hourly

Orientation #1

Fixed plane
Tilt/Azimuth 21 / -4 °

Sheds configuration
Nb. of sheds 9 units
Single array
Shading limit angle
Limit profile angle 31.3 °

Sizes
Sheds spacing 3.50 m
Sensitive width 2.28 m
GCR Shading 65.7 %
Top inactive band 0.02 m
Bottom inactive band 0.02 m

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Free Horizon

Near Shadings
According to strings : Fast (table)
Electrical effect 100 %

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module
Manufacturer Generic
Model HYPERSOL VSMDH.72.AAA.05
(Custom parameters definition)
HYPERSOL VSMDH.72.AAA.05.PAN
Unit Nom. Power 580 Wp
Number of PV modules 135 units
Nominal (STC) 78.3 kWp
Modules 9 units x 15 In series
At operating cond. (50°C)
Pmpp 72.9 kWp
U mpp 594 V
I mpp 123 A

Inverter
Manufacturer Generic
Model SH80CX
(Original PVsyst database)
Unit Nom. Power 80.0 kWac
Number of inverters 9 * MPPT 10% 0.9 unit
Total power 72.0 kWac
Operating voltage 180-1000 V
Max. power (=>30°C) 88.0 kWac
Pnom ratio (DC:AC) 1.09
No power sharing between MPPTs

Total PV power
Nominal (STC) 78 kWp
Total 135 modules
Module area 349 m²
Cell area 322 m²

Total inverter power
Total power 72 kWac
Nb. of inverters 1 unit
0.1 unused
Pnom ratio 1.09

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 79 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.75 %

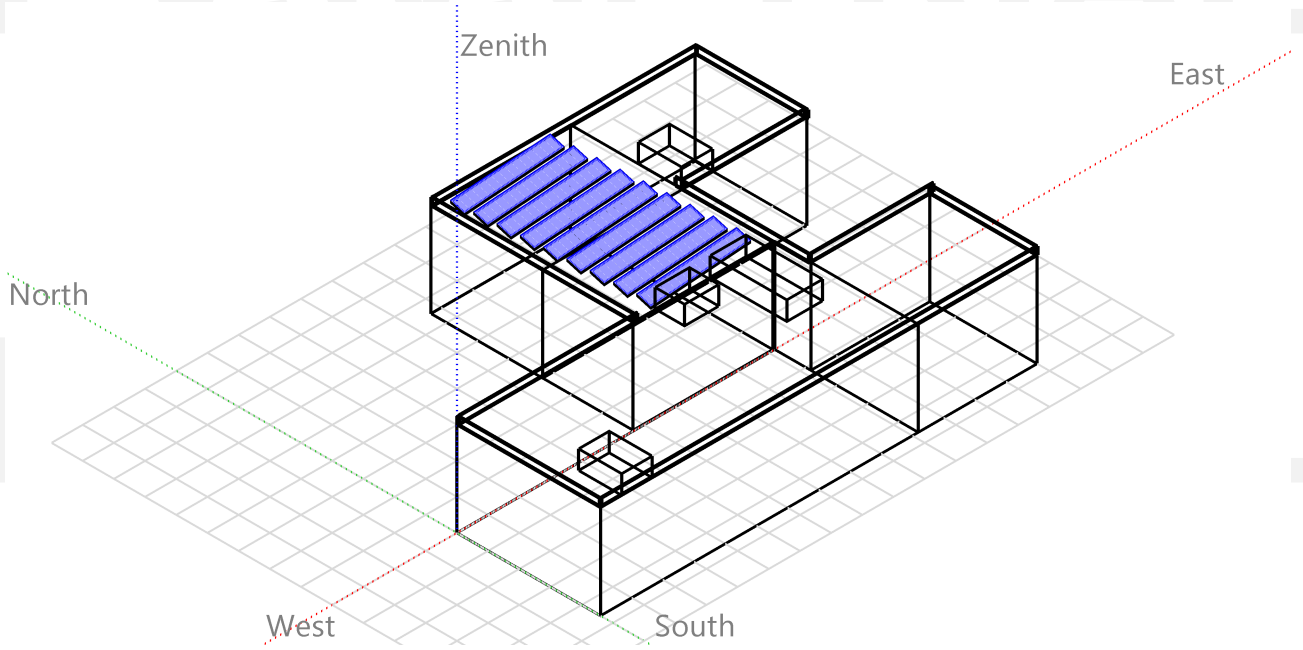
Module mismatch losses
Loss Fraction 2.00 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.402	0.000

Near shadings parameter

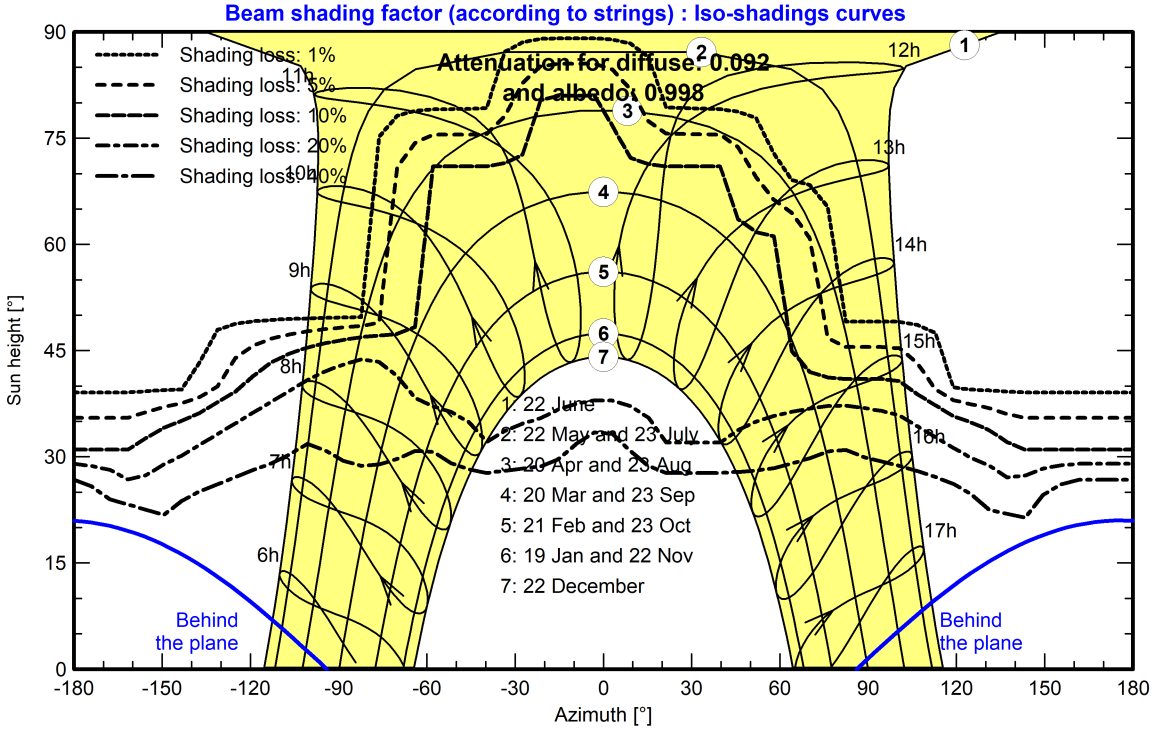
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 21°/ -4°

Beam shading factor (according to strings) : Iso-shadings curves



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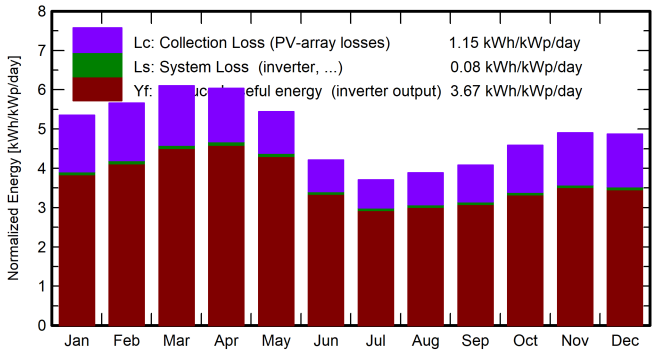
Main results

Simulation step
Hourly

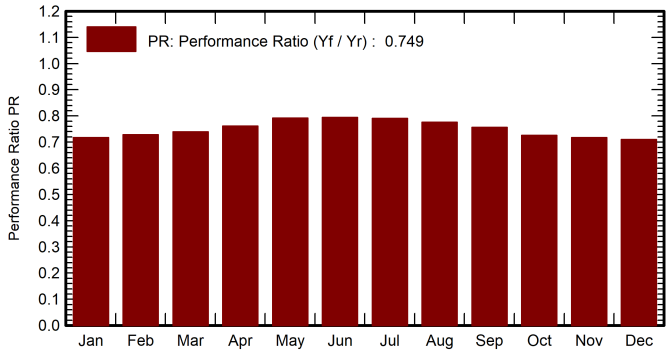
System Production

Produced Energy	104858 kWh/year	Specific production	1339 kWh/kWp/year
		Perf. Ratio PR	74.89 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	131.4	51.59	17.47	165.9	149.7	9508	9325	0.718
February	134.9	56.01	22.13	158.5	146.0	9213	9033	0.728
March	174.1	78.94	27.26	189.0	177.3	11159	10947	0.740
April	180.2	88.51	30.41	181.0	170.8	11005	10793	0.761
May	178.3	95.52	31.20	168.7	158.5	10662	10456	0.792
June	136.3	89.20	30.80	126.4	116.8	8028	7861	0.794
July	122.8	82.90	29.63	115.0	105.5	7280	7118	0.791
August	123.7	79.14	29.26	120.6	111.5	7484	7320	0.775
September	119.2	77.61	28.60	122.4	112.6	7416	7253	0.757
October	127.2	64.00	27.20	142.3	130.9	8253	8085	0.726
November	120.4	56.32	23.50	147.2	132.7	8427	8264	0.717
December	118.6	51.34	18.77	151.1	134.7	8573	8404	0.710
Year	1667.3	871.08	26.37	1788.2	1647.1	107008	104858	0.749

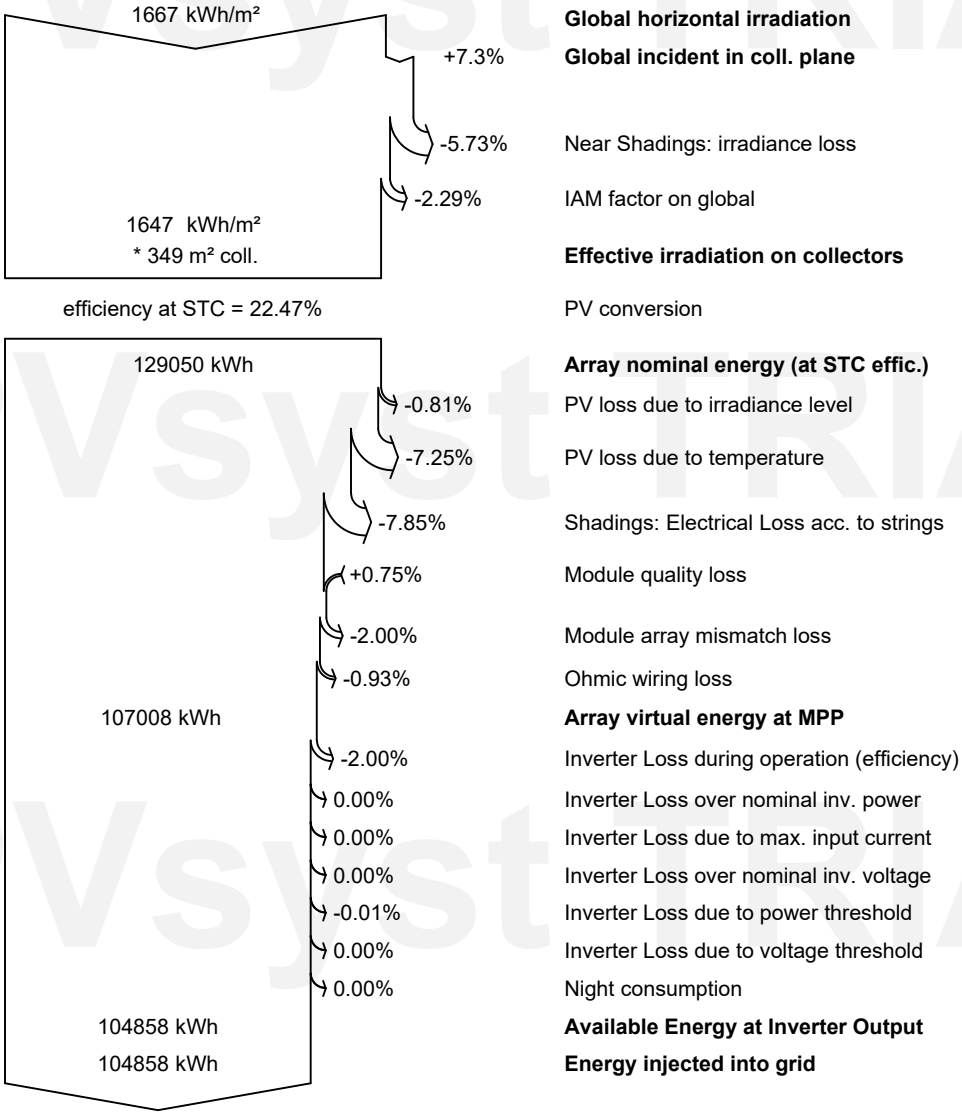
Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

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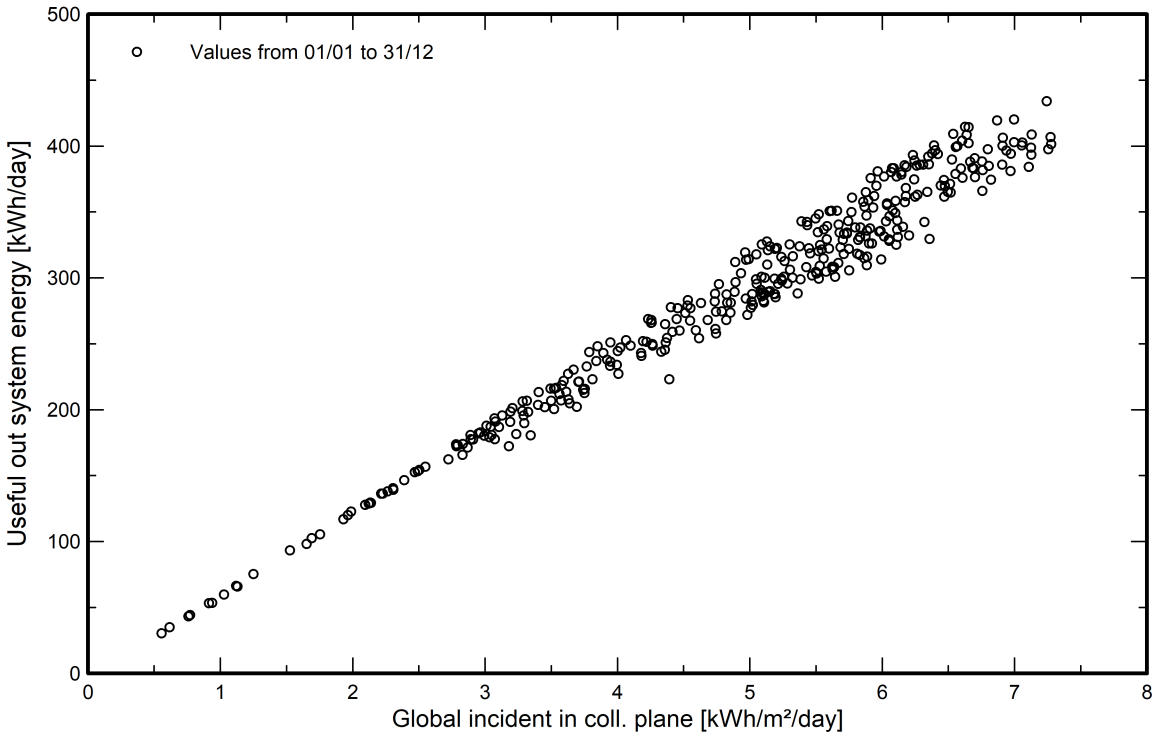
Loss diagram



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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

